1 Your number

Last 3 digits of ID: 121

$121 \geq 100 \rightarrow N = 121_{10}$

2. Decimal → Binary (Repeated Division by 2)

Division	Quotient	Remainder
$121 \div 2$	60	1
$60 \div 2$	30	0
$30 \div 2$	15	0
$15 \div 2$	7	1
$7 \div 2$	3	1
$3 \div 2$	1	1
$1 \div 2$	0	1

Bottom → top

$121_{10} = 1111001_2$

3 Decimal → OCTAL (Repeated division by 8)

Division	Quotient	Remainders
$121 \div 8$	15	1
$15 \div 8$	1	7
$1 \div 8$	0	1

Bottom → top: $121_{10} = 171_8$

4.

4. Decimal → Hexadecival (Repeated division by 16)

Division	Quotient	Remainder	Hex digit
$121 \div 16$	7	9	9
$7 \div 16$	0	7	7

Bottom top: $121_{10} = 79_{16}$

5. Verify - Convert back to Decimal

Binary → Decimal: 1111001_2

64 32 16 8 4 2 1

1 1 1 0 0 1

64 + 32 + 16 + 0 + 0 + 0 + 1 = 121

Octal → Decimal: 171_8

$1 \times 8^2 + 7 \times 8^1 + 1 \times 8^0$

$= 64 + 56 + 1 = 121$

Hex → Decimal: 79_{16}

$7 \times 16^1 + 9 \times 16^0$

$= 112 + 9 = 121$

6. Grouping shortcuts

Binary → Octal (group 3 from right)

1 111 001 → pad left → 001 111 001

→ 171 → 171_8

Binary → Hex (group by 4 from right)

1111 001 → pad left → 0111 1001 → 7 9 → 79_{16}

7. Error Analysis - 46_{10} → Binary

Division	Quotient	Remainder
46 ÷ 2	23	0
23 ÷ 2	11	1
11 ÷ 2	5	1
5 ÷ 2	2	1
2 ÷ 2	1	0
1 ÷ 2	0	1

Correct answer: 101110_2